Tinomutendayi Muzondidya

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Professional Summary

- Data Scientist and Industrial Engineer with expertise in optimisation modelling, NLP systems, and applied machine learning. Experienced in building end-to-end data pipelines, retrieval-augmented generation (RAG) systems, and decision-support tools using Python. Strong background in policy analytics, and responsible AI integration.

Technical Skills

- **Programming:** Python, R, SQL
- **Data Science & ML:** pandas, NumPy, scikit-learn, PyTorch, transformers, Sentence-Transformers, feature engineering, model evaluation, optimisation modelling
- **NLP & LLM Systems:** Retrieval-Augmented Generation (RAG), FAISS, semantic search, prompt engineering, embedding models
- **Data Visualisation:** Tableau, Matplotlib, Seaborn, Excel
- **Tools:** Git, GitHub, Streamlit, VS Code, API integration

Experience

- **Data Science & Policy Innovation Analyst** Apr 2024 – Feb 2026
  Policy Innovation Lab, Stellenbosch University *South Africa*
  - Designed Python-based data pipelines and research workflows supporting national-level projects on LLM adoption in South African governance.
  - Conducted comparative cloud infrastructure and regulatory analysis (US, EU, China), producing a policy briefing on data sovereignty and AI governance risks.
  - Cleaned and filtered a 2000+ entry EU Public Sector dataset using structured inclusion criteria, producing a curated 700+ AI tools catalogue relevant to government deployment.
  - Integrated Generative AI responsibly into research workflows, applying verification, bias mitigation, and governance-aligned documentation practices.
- **Sustainable Development Goals Consultant** Sep 2025 – Dec 2025
  Future17 Challenge (Remote)
  - Engineered **EcoDevice**, a React-based decision-support tool quantifying device lifecycle emissions and identifying circular economy interventions.
  - Built a carbon impact model incorporating manufacturing data, usage intensity, and regional grid emissions factors.

Projects

- **Fantasy Premier League Optimiser** 2025
  Python | Optimisation | API Modelling
  - Built a constrained optimisation model to construct an optimal 15-player squad under budget and positional constraints using real-time API data.
  - Engineered performance-based features to estimate projected player value and implemented expected-value based starting XI selection.
  - Developed a transfer optimisation module maximising marginal expected points gain under single-transfer constraints.
- **Personal Codex Agent** 2025
  Python | RAG | FAISS | Streamlit
  - Designed a RAG system using Sentence-Transformers embeddings and FAISS vector search for context-aware QA over curated documents.
  - Implemented chunking, indexing, cosine similarity retrieval, and OpenAI-based response generation with citation grounding.
  - Built a modular architecture (ingestion, retrieval, prompting) and deployed via Streamlit with configurable model parameters.

- **Artwork Classification with EfficientNet-B2** 2024
  PyTorch | Computer Vision | Deep Learning
  - Trained and fine-tuned an EfficientNet-B2 convolutional neural network for multi-class artwork classification using PyTorch and TorchVision.
  - Developed an end-to-end pipeline in Jupyter for data preparation, model training, evaluation, and prediction generation.
  - Implemented performance monitoring and model checkpointing, saving trained weights for reproducibility and deployment.
  - Generated structured prediction outputs for downstream evaluation and submission workflows.

Education

- **Stellenbosch University & University of Groningen** South Africa & the Netherlands
  Joint Doctoral Programme (PhD) *June 2026 – Present*
  - Research Focus: Modelling Latent Intent in Conversational Recommender Systems towards Accurate and Consistent Personalization.
  - Conducting the research through the Standard Bank Lab for Recommender Systems.
- **Stellenbosch University** South Africa
  Master of Engineering in Industrial Engineering (Data Science Focus) *Feb 2024 – March 2026*
  - Research Thesis (Result: 80%): Applied transformer models to analyze unstructured user feedback and improve recommender systems.
  - Academic Performance: Maintained a 77% average across Master of Engineering coursework, including Distinction in Big Data Technologies, Optimisation, and Deep Learning.
- **Stellenbosch University** South Africa
  BEng Chemical Engineering *Feb 2019 – Mar 2023*
  - Final Year Project: Steam distillation kinetics for D-limonene recovery from citrus waste.
  - Design Project: Preliminary design of a dual-pressure nitric acid production plant.
- **St. George's College** Zimbabwe
  Cambridge A-Level Certificate *Jan 2017 – Dec 2018*
  - Academic Excellence: Mathematics (A), Chemistry (A), Physics (B).
  - Awarded Full Colours for O-Level performance (9 A's).

Leadership & Community Service

- Book Lovers Book Club Coordinator (2024–Present)
- Treasurer, Maties Engineers Without Borders (2022).
- Head of Community Outreach, Stellenbosch Catholic Students Society (2019).
- Chess Captain & House Prefect, St. George's College (2018).
- Volunteer Teacher, Jairos Jiri Rehabilitation Centre (2017).

References

- **Prof. Willem Fourie**
  Professor, Chair of Policy Innovation and Policy Innovation Lab Lead
  willemf@sun.ac.za
- **Dr Sunday O. Oladejo**
  Chair of the Standard Bank Lab for Recommender Systems (SBLab)
  sunday@sun.ac.za